

Computer Networks - CS132/EECS148 - Spring 2013

Instructor: Karim El Defrawy

Assignment 5

Deadline : May 30th – 9:30pm (hard and soft copies required)

Problem 1 (Chapter 6 , problem 10 - 6 points) - Consider the following idealized LTE scenario: The downstream channel (see figure 6.20) is slotted in time, across F frequencies. There are four nodes A,B,C and D, reachable from a base station at rates of 10 Mbps, 5 Mbps, 2.5 Mbps and 1 Mbps , respectively, on the downstream channel. These rates assume that the base station utilizes all time slots available on all F frequencies to send to just one station. The base station has an infinite amount of data to send to each of the nodes, and can send to any one of these four nodes using any of the F frequencies during any timeslot in the downstream sub-frame.

- What is the maximum rate at which the base station can send to the nodes, assuming it can send to any node it chooses during each timeslot? Is your solution fair? Explain and define what you mean by “fair.”
- If there is a fairness requirement that each node must receive an equal amount of data during each one second interval, what is the average transmission rate by the base station (to all nodes) during the downstream sub-frame? Explain how you arrived at your answer.
- Suppose that the fairness criterion is that any node can receive at most twice as much data as any other node during the sub-frame. What is the average transmission rate by the base station (to all nodes) during the sub-frame? Explain how you arrived at your answer.

a. 10 Mbps if it only transmits to node A. This solution is not fair since only A is getting served. By “fair” it means that each of the four nodes should be allotted equal number of slots.

a. For the fairness requirement such that each node receives an equal amount of data during each downstream sub-frame, let n_1 , n_2 , n_3 , and n_4 respectively represent the number of slots that A, B, C and D get.

Now,

data transmitted to A in 1 slot = 10t Mbits
(assuming the duration of each slot to be t)

Hence,

Total amount of data transmitted to A (in n_1 slots) = 10t n_1

Similarly total amounts of data transmitted to B, C, and D equal to $5t n_2$, $2.5t n_3$, and $t n_4$ respectively.

Now, to fulfill the given fairness requirement, we have the following condition:

$$10t n_1 = 5t n_2 = 2.5t n_3 = t n_4$$

Hence,

$$n_2 = 2 n_1$$

$$n_3 = 4 n_1$$

$$n_4 = 10 n_1$$

Now, the total number of slots is N. Hence,

$$n_1 + n_2 + n_3 + n_4 = N$$

$$\text{i.e. } n_1 + 2 n_1 + 4 n_1 + 10 n_1 = N$$

$$\text{i.e. } n_1 = N/17$$

Hence,

$$n_2 = 2N/17$$

$$n_3 = 4N/17$$

$$n_4 = 10N/17$$

The average transmission rate is given by:

$$\begin{aligned} & (10t n_1 + 5t n_2 + 2.5t n_3 + t n_4) / tN \\ &= (10N/17 + 5 * 2N/17 + 2.5 * 4N/17 + 1 * 10N/17) / N \\ &= 40/17 = 2.35 \text{ Mbps} \end{aligned}$$

a. Let node A receives twice as much data as nodes B, C, and D during the sub-frame.

Hence,

$$10t n_1 = 2 * 5t n_2 = 2 * 2.5t n_3 = 2 * t n_4$$

$$\text{i.e. } n_2 = n_1$$

$$n_3 = 2n_1$$

$$n_4 = 5n_1$$

Again,

$$n_1 + n_2 + n_3 + n_4 = N$$

$$\text{i.e. } n_1 + n_1 + 2n_1 + 5n_1 = N$$

$$\text{i.e. } n_1 = N/9$$

Now, average transmission rate is given by:

$$\begin{aligned} & (10t n_1 + 5t n_2 + 2.5t n_3 + t n_4) / tN \\ &= 25/9 = 2.78 \text{ Mbps} \end{aligned}$$

Similarly, considering nodes B, C, or D receive twice as much data as any other nodes, different values for the average transmission rate can be calculated.

Problem 2 (Chapter 6 , problem 13 - 3 points) - In mobile IP, what effect will mobility have on end-to-end delays of datagrams between the source and destination.

Because datagrams must be first forward to the home agent, and from there to the mobile, the delays will generally be longer than via direct routing. Note that it is possible, however, that the direct delay from the correspondent to the mobile (i.e., if the datagram is not routed through the home agent) could actually be smaller than the sum of the delay from the correspondent to the home agent and from there to the mobile. It would depend on the delays on these various path segments. Note that indirect routing also adds a home agent processing (e.g., encapsulation) delay.

Problem 3 (Chapter 6 , problem 15 - 3 points) - Consider two mobile nodes in a foreign network having a foreign agent. Is it possible for two mobile nodes to use the same care-of in mobile IP? Explain your answer.

Two mobiles could certainly have the same care-of-address in the same visited network. Indeed, if the care-of-address is the address of the foreign agent, then this address would be the same. Once the foreign agent decapsulates the tunneled datagram and determines the address of the mobile, then separate addresses would need to be used to send the datagrams separately to their different destinations (mobiles) within the visited network.

Problem 4 (Chapter 8 , problem 4 - 6 points) - Consider the block cipher in Figure 8.5. Suppose that each block cipher T_i simply reverses the order of the 8 input bits (so that, for example, 11110000 becomes 00001111). Further suppose that the 64 bit scrambler does not modify any bits (so that the output value of the m th bit is equal to the input value of the m th bit). (a) With $n = 3$ and the original 64 bit input equal to 10100000 repeated 8 times, what is the value of the output? (b) Repeat part a but now change the last bit of the original 64 bit input from a 0 to a 1. (c) Repeat parts a and b but now suppose that the 64 bit scrambler inverses the order of the 64 bits.

- a. The output is equal to 00000101 repeated eight times.
- b. The output is equal to 00000101 repeated seven times + 10000101.
- c. We have $(ARBR)R = CBA$, where A, B, C are strings, and R means inverse operation. Thus:
 1. For (a), the output is 10100000 repeated eight times;
 2. For (b), the output is 10100001 + 10100000 repeated seven times.

Problem 5 (Chapter 8 , problem 8 - 6 points) - Consider RSA with $p=5$ and $q=11$.

- What are n and z ?
- Let e be 3. Why is this an acceptable choice for e ?
- Find d such that $de=1 \pmod{z}$ and $d < 160$.
- Encrypt the message $m=8$ using the key (n,e) . Let c denote the corresponding ciphertext. Show all work. Hint: To simplify the calculations use the fact :
 $[(a \bmod n) \cdot (b \bmod n)] \bmod n = (a \cdot b) \bmod n$

$p = 5, q = 11$

- $n = p \cdot q = 55, z = (p-1)(q-1) = 40$
- $e = 3$ is less than n and has no common factors with z .
- $d = 27$
- $m = 8, me = 512, \text{Ciphertext } c = me \bmod n = 17$

Problem 6 (Chapter 8 , problem 9 - 6 points) - In this problem we explore the Diffie-Hellman (DH) public-key encryption algorithm, which allows two entities to agree on a shared key. The DH algorithm makes use of a large prime number p and another large number g less than p . Both p and g are made public (so that an attacker would know them). In DH, Alice and Bob each independently choose secret keys, S_A and S_B respectively. Alice then computes her public key, T_A , by raising g to S_A and then taking mod p . Bob similarly computes his own public key T_B by raising g to S_B and then taking mod p . Alice and Bob then exchange their public keys over the internet. Alice then calculates the shared secret key S by raising T_B to S_A and then taking mod p . Similarly Bob calculates shared key S' by raising T_A to S_B and then taking mod p .

- Prove that, in general, Alice and Bob obtain the same symmetric key, that is, prove $S = S'$.
- With $p=11$ and $g=2$, suppose Alice and Bob choose private keys $S_A=5$ and $S_B = 12$, respectively. Calculate Alice's and Bob's public keys, T_A and T_B . Show all work.
- Following up on part b, now calculate S as the shared symmetric key. Show all work.
- Provide a timing diagram that shows how DH can be attacked by a man-in-the-middle. The timing diagram should have three vertical lines, one for Alice, one for Bob, one for the attacker Trudy.

<u>Alice</u>	<u>Bob</u>
secret key: S_A	S_B
public key: $T_A = (g^{S_A}) \bmod p$	$T_B = (g^{S_B}) \bmod p$
shared key: $S = (T_B^{S_A}) \bmod p$	$S' = (T_A^{S_B}) \bmod p$

$$\begin{aligned} \text{a. } S &= (T_B^{S_A}) \bmod p = ((g^{S_B} \bmod p)^{S_A}) \bmod p = (g^{(S_B S_A)}) \bmod p \\ &= ((g^{S_A} \bmod p)^{S_B}) \bmod p = (T_A^{S_B}) \bmod p = S' \end{aligned}$$

(b and c) $p = 11, g = 2$

	<u>Alice</u>	<u>Bob</u>
secret key:	$S_A = 5$	$S_B = 12$
public key:	$T_A = (g^{S_A}) \bmod p = 10$	$T_B = (g^{S_B}) \bmod p = 4$
shared key:	$S = (T_B^{S_A}) \bmod p = 1$	$S' = (T_A^{S_B}) \bmod p = 1$

d)

T_A

T_T

)

T_T

T_B

Alice

Trudy

Bob

The Diffie-Hellman public key encryption algorithm is possible to be attacked by man-in-the-middle.

1. In this attack, Trudy receives Alice's public value (T_A) and sends her own public value (T_T) to Bob.
2. When Bob transmits his public value (T_B), Trudy sends her public key to Alice (T_T).
3. Trudy and Alice thus agree on one shared key (S_{AT}) and Trudy and Bob agree on another shared key (S_{BT}).

After this exchange, Trudy simply decrypts any messages sent out by Alice or Bob by the public keys S_{AT} and S_{BT} .

Problem 7 (Chapter 8 , problem 15 - 6 points) - Consider our authentication protocol in Figure 8.18, in which Alice authenticates herself to Bob, which we saw works well (i.e. we found no flaws in it). Now suppose that while Alice is authenticating herself to Bob, Bob must authenticate himself to Alice. Give a scenario by which Trudy, pretending to be Alice, can now authenticate herself to Bob as Alice. (Hint: consider that the sequence of operations of the protocol, one with Trudy initiating and one with Bob initiating, can be arbitrarily interleaved. Pay particular attention to the fact that both Bob and Alice will use a nonce, and that if care is not taken, the same nonce can be used maliciously)

Bob does not know if he is talking to Trudy or Alice initially. Bob and Alice share a secret key K_{A-B} that is unknown to Trudy. Trudy wants Bob to authenticate her (Trudy) as Alice. Trudy is going to have Bob authenticate himself, and waits for Bob to start:

1. **Bob-to-Trudy: "I am Bob"** Commentary: Bob starts to authenticate himself. Bob's authentication of himself to the other side then stops for a few steps.
2. **Trudy-to-Bob: "I am Alice"** Commentary: Trudy starts to authenticate herself as Alice
3. **Bob-to-Trudy: "R"** Commentary: Bob responds to step 2 by sending a nonce in reply. Trudy does not yet know $K_{A-B}(R)$ so she can not yet reply.
4. **Trudy-to-Bob: "R"** Commentary: Trudy responds to step 1 now continuing Bob's authentication, picking as the nonce for Bob to encrypt, *the exact same value that Bob sent her to encrypt in Step 3.*
5. **Bob-to-Trudy: " $K_{A-B}(R)$ "** Bob completes his own authentication of himself to the other side by encrypting the nonce he was sent in step 4. Trudy now has $K_{A-B}(R)$. (Note: she does not have, nor need, K_{A-B})

Trudy-to-Bob: " $K_{A-B}(R)$ " Trudy completes her authentication, responding to the R that Bob sent in step 3 above with $K_{A-B}(R)$. Since Trudy has returned the properly encrypted nonce that Bob sent in step 3, Bob thinks Trudy is Alice!

Problem 8 (Chapter 8 , problem 21 - 4 points) - Suppose Alice and Bob are communicating over an SSL session. Suppose an attacker, who does not have any of the shared keys, inserts a bogus TCP segment into a packet stream with correct TCP checksum and sequence numbers (and correct IP addresses and port numbers). Will SSL at the receiving side accept the bogus packet and pass the payload to the receiving application? Why or why not?

No, the bogus packet will fail the integrity check (which uses a shared MAC key).

Problem 9 (Chapter 8 , problem 22 - 8 points) - The following True/False questions pertain to Figure 8.28 of your book.

- a. When a host in 172.16.1/24 sends a datagram to an Amazon.com server, the router R1 will encrypt the datagram using IPsec.

b. When a host in 172.16.1/24 sends a datagram to a host in 172.16.2/24, the router R1 will change the source and destination address of the IP datagram.

c. Suppose a host in 172.16.1/24 initiates a TCP connection to a Web server in 172.16.2/24. As part of this connection, all datagrams sent by R1 will have protocol number 50 in the left-most IPv4 header field.

d. Consider sending a TCP segment from a host in 172.16.1/24 to a host in 172.16.2/24. Suppose the ACK for this segment gets lost, so that TCP resends the segment. Because IPsec uses sequence numbers, R1 will not resend the TCP segment.

a) F

b) T

c) T

d) F

Problem 10 (Chapter 8 , problem 24 - 8 points) - Consider the following pseudo-WEP protocol. The key is 4 bits and the IV is 2 bits. The IV is appended to the end of the key when generating the keystream. Suppose that the shared secret key is 1010. The keystreams for the four possible inputs are as follows:

101000: 00101011101010101001011010100100 ...

101001: 1010011011001010110100100101101 ...

101010: 0001101000111100010100101001111 ...

101011: 1111101010000000101010100010111 ...

Suppose all messages are 8 bits long. Suppose ICV (integrity check) is 4 bits long, and is calculated by XORing the first 4 bits of the data with the last 4 bits of the data. Suppose the pseudo-WEP packet consists of three fields: first the IV field, then the message field and last the ICV field, with some of these fields encrypted.

a. We want to send a message $m=10100000$ using the $IV = 11$ and using WEP. What will be the values in the three WEB fields ?

b. Show that when the receiver decrypts the WEP packet, it recovers the message and the ICV.

c. Suppose Trudy intercepts a WEP packet (not necessarily with $IV = 11$) and wants to modify it before forwarding to the receiver. Suppose Trudy flips the first ICV bit. Assuming that Trudy does not know the keystreams for any of the IVs, what other bit(s) must Trudy also flip so that the received packet passes the ICV check?

d. Justify your answer by modifying the bits in the WEP packet in part (a), decrypting the resulting packet, and verifying the integrity check.

a. Since IV = 11, the key stream is 111110100000

Given, $m = 10100000$

Hence, $ICV = 1010 \text{ XOR } 0000 = 1010$

The three fields will be:

IV: 11

Encrypted message: $10100000 \text{ XOR } 11111010 = 01011010$

Encrypted ICV: $1010 \text{ XOR } 0000 = 1010$

a. The receiver extracts the IV (11) and generates the key stream 111110100000

XORs the encrypted message with the key stream to recover the original message:

$01011010 \text{ XOR } 11111010 = 10100000$

XORs the encrypted ICV with the keystream to recover the original ICV:

$1010 \text{ XOR } 0000 = 1010$

The receiver then XORs the first 4 bits of recovered message with its last 4 bits:

$1010 \text{ XOR } 0000 = 1010$ (which equals the recovered ICV)

a. Since the ICV is calculated as the XOR of first 4 bits of message with last 4 bits of message, either the 1st bit or the 5th bit of the message has to be flipped for the received packet to pass the ICV check.

a. For part (a), the encrypted message was 01011010

Flipping the 1st bit gives, 11011010

Trudy XORs this message with the keystream:

$11011010 \text{ XOR } 11111010 = 00100000$

If Trudy flipped the first bit of the encrypted ICV, the ICV value received by the receiver is 0010

The receiver XORs this value with the keystream to get the ICV:

$0010 \text{ XOR } 0000 = 0010$

The receiver now calculates the ICV from the recovered message:

$0010 \text{ XOR } 0000 = 0010$ (which equals the recovered ICV and so the received packet passes the ICV check)

Problem 11 (Chapter 8 , problem 25 - 4 points) - Provide a filter table and a connection table for a stateful firewall that is as restrictive as possible but accomplished the following:

- a. Allows all internal users to establish Telnet sessions with external hosts.
- b. Allows external users to surf the company Web site at 222.22.0.12.
- c. But otherwise blocks all inbound and outbound traffic.

The internal network is 222.22/16. In your solution, suppose that the connection table is currently caching three connections, all from inside to outside. You'll need to invent appropriate IP addresses and port numbers.

Filter Table:

Action	Source Address	Dest address	Protocol	Source port	Dest port	Flag bit	Check connection
allow	222.22/16	outside of 222.22/16	TCP	> 1023	23	any	
allow	outside of 222.22/16	222.22/16	TCP	23	> 1023	ACK	x
Allow	outside of 222.22/16	222.22.0.12	TCP	>1023	80	Any	
Allow	222.22.0.12	outside of 222.22/16	TCP	80	>1023	Any	
deny	All	all	all	all	all	All	

Connection Table:

Source address	Dest address	Source port	Dest port
222.22.1.7	37.96.87.123	12699	23
222.22.93.2	199.1.205.23	37654	23
222.22.65.143	203.77.240.43	48712	23